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Zero-Shot Strike: Testing the generalisation capabilities of out-of-the-box LLM models for depression detection

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ABSTRACT

Depression is a significant global health challenge. Still, many people suffering from depression remain undiagnosed. Furthermore, the assessment of depression can be subject to human bias. Natural Language Processing (NLP) models offer a promising solution. We investigated the potential of four NLP models (BERT, Llama2-13B, GPT-3.5, and GPT-4) for depression detection in clinical interviews. Participants (N = 82) underwent clinical interviews and completed a self-report depression questionnaire. NLP models inferred depression scores from interview transcripts. Questionnaire cut-off values for depression were used as a classifier for depression. GPT-4 showed the highest accuracy for depression classification (F1 score 0.73), while zero-shot GPT-3.5 initially performed with low accuracy (0.34), improved to 0.82 after fine-tuning, and achieved 0.68 with clustered data. GPT-4 estimates of symptom severity PHQ-8 score correlated strongly (r = 0.71) with true symptom severity. These findings demonstrate the potential of AI models for depression detection. However, further research is necessary before widespread deployment can be considered.

1. Introduction

According to the World Health Organization (2023), 280 million people worldwide are currently suffering from depression, with rising trends evident in meta-analyses (Moreno-Agostino et al., 2021). While Lim et al. (2018) found a pooled point-prevalence of 15.4% for depression in studies published from 2004–2014, a more recent meta-analysis (Bueno-Notivol et al., 2021) found a pooled prevalence of 25% during the COVID-19 outbreak. Prior to the COVID-19 pandemic, an increase in major depression was reported, particularly among US adolescents and young adults (Mojtabai et al., 2016). Contributing factors to the rise in depression include the 2008 economic crisis and following Great Recession, leading to increased financial insecurity and inequality (Martin-Carrasco et al., 2016), as well as heightened awareness and reporting of depression in the community, partly due to the availability of online mental health resources (Lim et al., 2018). During the COVID-19 pandemic, increased depression rates can be attributed to various factors, including lockdown measures, uncertainty, home confinement, quarantine, socioeconomic losses such as unemployment,

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lack of social support, and ongoing concerns about personal and others' well-being (Bueno-Notivol et al., 2021; Daniali et al., 2023; Mahmud et al., 2023). Despite depression being the leading course of disability worldwide (Friedrich, 2017), some evidence points to a potential underdiagnosis (Fekadu et al., 2022; Sheehan, 2004; Wancata and Friedrich, 2011). Various factors, such as age (Allan et al., 2014; Sigalas et al., 2014), race and ethnicity (Shao et al., 2016; Wyman et al., 2020) or country income (Fekadu et al., 2022) have been found to contribute to depression underdiagnosis. This is concerning, as depression negatively impacts the quality of life of patients (Hohls et al., 2021) and their immediate families (Radfar et al., 2014), increases the risk of mortality (Mykletun et al., 2007) and poses a significant economic burden (Greenberg et al., 2021). Still, the number of people receiving minimally adequate treatment is alarmingly low, with detection rates of around 20% in high-income countries and 3.7% in countries with lower income (Thornicroft et al., 2017). Regarding these numbers, it is important to notice cultural differences and their potential contribution to depression rates: For example, depression may be perceived primarily as somatic in non-Western countries, diagnostic tools may lack validation for some countries and cultural groups, which might influence prevalences (Thornicroft et al., 2017), and diagnostic thresholds vary across countries (Kessler and Bromet, 2013). Research by Krendl and Pescosolido (2020) indicates higher stigma levels and greater moral attributions associated with depression in Eastern countries, which might lead to an underreporting of depression.

There are multiple barriers to attaining a depression diagnosis, such as stigma (Arnaez et al., 2020), lack of resources for treatment referrals, and lack of time during medical encounters (Colligan et al., 2020). Furthermore, there is evidence that depression screening in primary care is negatively impacted by biases for culture, gender, ethnicity/race, income, and sexual orientation (Kerr and Kerr, 2001; Levkovich and Elyoseph, 2023; Riley and Hill-Daniel, 2020; Sha and Aleshire, 2023). As assessment for depression is mostly done in clinical interviews, Large Language Models (LLM) offer the potential to overcome depression assessment barriers, providing easily accessible solutions with the potential for an unbiased perspective. Natural Language Processing (NLP) models in particular show promise in making inferences from human language.

As presented in this paper, our approach leverages cutting-edge advancements and emerging trends in Natural Language Processing (NLP), utilising state-of-the-art Large Language Models (LLMs) for detecting depression from a newly collected dataset of transcribed semi-clinical depression interviews. These comparisons are conducted for fine-tuned and zero-shot models. Furthermore, the models are tasked with an evaluation of depression severity and the impact of a clustering technique to improve contextual comprehension is explored. The test dataset (further referred to as KID dataset) is made up of interviews between participants from general population and a trained team of clinical psychology students. All interviews are recorded, transcribed, translated and processed, following uniform procedure ensuring comparability and then used to test efficacy of different LLMs in depression detection from textual input.

2. Related work

Given the prevalence and negative impact of depression, it is not surprising that research on NLP models for mental health diagnostics mostly focuses on the detection of this disorder (Zhang et al., 2022). As a study by Wolohan et al. (2018) demonstrated, linguistic markers of depression can be found by Linguistic Inquiry and Word Count (LIWC) analysis. Such features can be extracted by Deep Learning Models and enhanced by including contextual factors (Zhang et al., 2022). One such Deep Learning model and a significant advancement within the field of NLP models has been the Bidirectional Encoder Representations from Transformers (BERT) architecture by Devlin et al. (2019). This model has been used to detect vocabulary evidence of depression in clinical interview settings by Villatoro-Tello et al. (2021), who refer to Lexical Availability Theory (Gougenheim, 1964) as an inspiration for their approach. Further evidence for the diagnostic capabilities of BERT-based models has been documented by Rodrigues Makiuchi et al. (2019) and Senn et al. (2022). A further milestone within the development of NLP models has been the introduction of OpenAI's GPT-3.5 model. The model has since been investigated for its capabilities in detecting depression from social media posts (Lamichhane, 2023; Qin et al., 2023) or transcribed semi-clinical interviews (Sadeghi et al., 2023). GPT-3.5 has since been outdated by its successor, GPT-4 (Rahaman et al., 2023). A study by Danner et al. (2023) showcases the diagnostic performance of GPT-4 on a preliminary dataset. While the body of research in this field is advanced with each new study, it is important to contemplate some key aspects in order to make sound inferences on the applicability of NLP models for depression detection. One such aspect is the criterion for depression: Only in a minority of studies included in a literature review by Ahmed et al. (2022) ($n = 9$ from $N = 54$) objective criteria, such as validated questionnaires, were used to categorise participants as "depressed". A review article by Zhang et al. (2022), which takes into consideration mental health conditions beyond depression, comes to the same conclusion: The majority of data sources within this field are social media posts and the criterion for "mental illness" differs, e.g. posting on subReddits for people with depression (Zhang et al., 2022).

Findings from such sources might also not transfer to the general population due to potential selection bias in an online sample. To ensure the effectiveness of a screening method across different age groups, further studies on a more generalised sample are essential. Additionally, if self-labelling as "depressed" or visiting spaces for depressed individuals are criteria, there is a risk of overlooking individuals with mild to moderate depressive symptoms or limited mental health literacy.

Given linguistic variations between written and spoken language (Leech, 1992), it is crucial to validate NLP model performance on interview data. (Zhang et al., 2022) note the limited use of interview data for mental illness inferences with AI and report that most studies primarily rely on the Distress Analysis Interview Corpus (DAIC) from the University of California (UCLA) (Gratch et al., 2014). While this specific dataset has also been used to train our BERT model, we generated a novel set of interview data including individuals from the general population. Furthermore, our study endeavours to directly compare the diagnostic capabilities of a fine-tuned BERT-base model with three zero-shot trained models, GPT-3.5, GPT-4 and Llama2-13B. We aim to fill the research gap concerning the feasibility of depression detection from clinical interviews within the general population with special regard to different NLP-models, such as GPT-4.

3. Methodology

3.1. Experimental design

The experimental design for our current investigation is in line with Hadžić et al. (2024), in which we ran a pilot study on a small dataset from our interview corpus (data from the pilot study are not included in the current study). A cross-sectional approach was chosen to assess and compare the procedural feasibility of using NLP models to detect depression in transcribed clinical interviews in the general population. Furthermore, we wanted our results to be comparable with other studies within this field of research (Lamichhane, 2023; Rodrigues Makiuchi et al., 2019; Sadeghi et al., 2023; Zhang et al., 2022), seeking to validate the generalisability of accuracies obtained within studies focusing on social media posts or obtained from the Distress Analysis Interview Corpus (DAIC) (Gratch et al., 2014). This design was used to test different NLP models for diagnosing depression.

3.2. Text classification models

In our experimentation, we chose three language models, namely BERT, GPT, and Llama. BERT was selected for its state-of-the-art status, GPT for its current state-of-the-art performance, and Llama due to its recognition as the most promising open-source option available (Touvron et al., 2023).

3.2.1. BERT

When introduced by a team of Google researchers (Devlin et al., 2019), Bidirectional Encoder Representations from Transformers (BERT), has emerged as a pivotal advancement in the field of Natural Language Processing (NLP). BERT's bidirectional processing, large-scale pre-training on diverse corpora, and contextual embeddings revolutionised language understanding, leading to state-of-the-art performance across various NLP tasks. Due to their unidirectional processing, traditional language models have limited pre-training options in contemporary architectures and therefore, this restriction presented a big problem that had to be solved (Shreyashree et al., 2022). In opposition to traditional unified language model pre-training, BERT takes into account a word's meaning on both sides across every layer of its neural network, which enables it to extract more contextual information from unstructured text (Devlin et al., 2019). BERT employs the Transformer architecture, which utilises self-attention mechanisms to weigh the importance of different words in the input sequence when encoding representations. This attention mechanism enables BERT to capture long-range dependencies more effectively compared to traditional sequential models like LSTMs or GRUs (Koroteev, 2021). BERT is pre-trained using two unsupervised tasks: masked language modeling (MLM) and next sentence prediction (NSP). This pre-training process allows BERT to learn rich, generalised representations of language. As a result, the pre-trained BERT model can be fine-tuned with just one additional output layer to create state-of-the-art models for a wide range of tasks, such as question answering and language inference, without substantial task specific architecture modifications (Devlin et al., 2019). The original BERT model, has two main variants: BERT base, including 110 million parameters and BERT large, based on 340 million parameters. BERT, as a base model, can then be trained for specific tasks using the task specific dataset and gradient descent algorithm to optimise the model weights. This step is called fine-tuning and is used to improve the model's application/task specific accuracy (Tribes et al., 2023). Since its introduction, BERT has been fine-tuned for various sets of specific tasks, domains, and purposes. As shown in Section 2 it has also been proven to be quite efficient in the field of depression detection. For the same purpose of depression detection on the text-based data, in our approach, we also fine-tuned the BERT base model with the goal of depression detection from textual data. A description of the training dataset is shown in Section 3.2.2. We fine-tuned our BERT base model on an NVIDIA GPU. With 12 hidden layers and 768 neurons per layer, our BERT classifier can handle two classes of classification tasks. The model included around 110 million parameters overall. Utilising the AdamW optimiser with a learning rate of 3×10^{-5}, training was carried out with a batch size of 16 across 15 epochs. Dropout rates were added, with 0.3 for hidden layers and 0.5 for attention layers. Furthermore, a 4×10^{-2} weight-decay was used. After conducting tuning runs with various combinations, these optimal hyper-parameters values were selected that yield the best performance. As the output, the model produces logit probability scores for both classes, while the binary option with a higher probability score between the two is selected as the final score. So given binary outputs of 0, meaning no depression predicted, and 1, with predicting depression, is capable of matching and comparing the scores with the clinical interview scores, where the scores are also defined as binary output, depressed or non-depressed.

3.2.2. Training dataset

The dataset for fine-tuning our BERT model came from the DAIC-WOZ and Extended DAIC dataset, both released by the University of Southern California - Institute for Creative Technologies (USCICT) as a part of the Distress Analysis Interview Corpus (DAIC) database (Gratch et al., 2014). These datasets have been used by studies researching the capabilities of NLP models in the detection of mental disorders worldwide (e.g. Ringeval et al., 2017). Collected during the development of a digital agent for mental disorder detection, the DAIC-WOZ dataset consists of 193 records (audio and video) in which trained interviewers interacted with participants by proxy of a virtual avatar the participants deemed to act autonomously (Wizard of Oz paradigm). In contrast to this, the 263 interviews within the extended DAIC dataset were carried out by the automated virtual agent 'Ellie'. This led to a variation in interview duration, with DAIC-WOZ interviews ranging from 5 to 20 min and Extended DAIC interviews spanning 15 to 25 minutes (Gratch et al., 2014). Participants within both DAIC datasets were either drawn from the general population or veterans of the U.S. Armed Forces. Among other mental health screening questionnaires, the PHQ-8 questionnaire (Kroenke et al., 2001) was filled out by the participants. This value was used as the 'ground truth' for our BERT model to determine intensity of depression. The model was trained with transcripts of the interview audio data.

3.2.3. GPT-3.5-turbo and GPT-4

The abbreviation GPT stands for “Generative Pre-trained Transformer,” and represents a family of LLMs developed by OpenAI. GPT models are trained using unsupervised learning on a large corpus of unlabelled data collected from various sources available on the internet. As its name suggests, GPT is a type of generative language model, meaning it is capable of generating content that is both coherent and relevant to the situation based on the input it receives from the user (Kalyan, 2023). The initial model was introduced by OpenAI in 2018, (Radford et al., 2018), one year later came the second version of the model (Radford et al., 2019), and in 2020 came the model 3 that caught worldwide attention as one of the first mass-used AI tools for everyday use. With 175 billion parameters, GPT-3 was the biggest LLM to date (Hadi et al., 2023). Versions 3.5 and 4 were trained on an even bigger, unprecedented amount of parameters and in literature are shown to achieve impressive results in various application fields.

These models are pre-trained on a large corpus of textual data from internet sources (e.g. books and articles) to grasp the structure and patterns of natural language. Following this, they undergo fine-tuning for specific tasks like translation or summary. This phase involves adjusting the model’s parameters to match the intricacies of the task, while retaining its general language understanding (Radford et al., 2019). GPT’s standout feature is its capability to generate coherent and contextually fitting text, making it proficient across diverse applications, even with zero or few shot training (Savelka, 2023; Bommarito et al., 2023). However, there is still a significant literature gap that this study is attempting to bridge in terms of academic research on GPT’s capacity to identify depression from text-based inputs.

3.2.4. GPT4All

GPT4ALL is a software ecosystem created and maintained by Nomic AI. Many open-source LLMs can easily be trained and deployed locally on consumer-grade CPUs using the GPT4All ecosystem. Many open-source LLM models such as GPT-J, Nours-Hermes, Llama etc., are available in their model hub. For our preliminary testing, we used the Llama2 model (Touvron et al., 2023) as currently most promising open source LLM. However, other available open-source models will also be tested in the future studies. Llama2 is an auto-regressive LLM developed and fine-tuned by Facebook, pre-trained using 2 trillion tokens of data from sources that are open to the public (Nguyen et al., 2023). In addition to over a million additional instances with human annotations, the fine-tuning data comprises publicly accessible instruction datasets. Llama2 comes in a range of parameter sizes — 7B, 13B, and 70B with a context length 4k tokens. Our computational setup features an AMD Ryzen 5 5600X processor, with 32 GB RAM. Therefore, we used the 13B parameters model (Llama2-13B) (Touvron et al., 2023) as it fit our memory requirements and offered reasoning and reading comprehension equal to the bigger model.

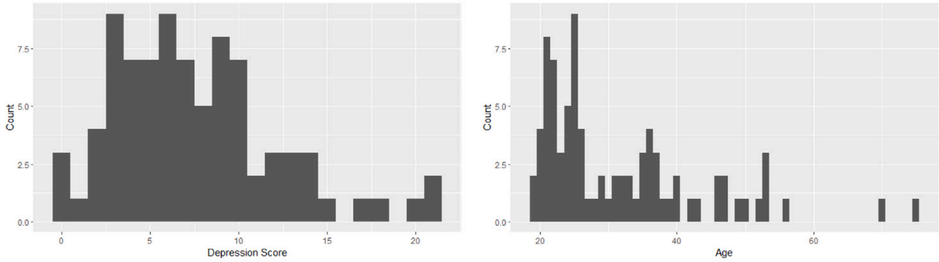
The utilisation of open-source and offline Language Models offers both advantages and disadvantages: Managing sensitive data on campus reduces the risk of data leaks and improves data protection and regulatory issues. This advantage of open-source models is especially emphasised when dealing with sensitive data, concerning mental health and depression. Such possibility to run the whole process offline and locally elevates the data protection aspects to even higher level of security. These models also ensure stability and reliability of their outputs through the time, unlike closed-source models that may yield varying responses with updates (Chen et al., 2023). Such reliability and stability are particularly crucial in the research domain, where model outputs must grant the possibility of results replication and comparability. Additionally, open-source model offer more options for customisation and fine-tuning process, while in the same time granting higher level of transparency. Open-source models provide user with the opportunity to inspect model architecture, training dataset as well as code, enhancing trust and transparency (Liu et al., 2023). However, open-source models may not be as up-to-date; for instance, the Llama2 model (Touvron et al., 2023) has a knowledge cut-off of September 2022, nearly a year behind compared to more frequently updated GPT-based models. Furthermore, updating and maintaining the model can demand a lot of time and resources (Liu et al., 2023). Concerns with intellectual property (IP) also arise when using open-source LLMs, particularly if the model was trained using sensitive or confidential data. Therefore, there are inherent legal and responsibility risks associated with developing commercial products, including open-source LLM. On the other hand, closed-source LLMs typically have explicit terms of service and legal agreements.

3.3. Participants

Participants were recruited from the German general population via postings on various social media platforms (X, Instagram, Facebook), via personal contacts of interviewers as well as the Course Credit board at the Private University of Applied Sciences Göttingen. Inclusion criteria were an age of 18 years and older and the ability to comprehend questionnaires in German. It was ensured that participants were not interviewed by individuals within their social circles. Participants completing the interview were provided with general information on depression as well as resources for therapy, including emergency phone numbers. The exclusion criterion was not completing the questionnaire. Before participation, participants were informed about the study and the recording of the interview and gave their consent according to the Declaration of Helsinki.

A still-growing dataset of currently $N = 200$ interviews was generated following this procedure. For this study, a sample of $N = 84$ participants was randomly drawn from this interview corpus. From this sample, n = 1 interview had to be removed due to technical issues during transcription. Within the sample, n = 65 (78.31%) participants identified as female, n = 16 (19.28%) participants identified as male and n = 2 (2.41%) participants identified as of diverse gender identity. Participant ages ranged from 19 to 75 years old, with a mean score of M = 31.64 (SD = 11.81). Age distribution through a sample is visually presented in Fig. 1(b). As can be seen, 29.27% of the sample scored 10 or higher on the PHQ-8, indicating depression (Kroenke et al., 2001). The average PHQ-8 score was M = 7.56 (SD = 4.67). The distribution of the PHQ-8 scores on the sample size is also illustrated in Fig. 1(a).

Consent to participate in the study and for voice recording following the Declaration of Helsinki was obtained from all participants. The Ethics Committee of the Private University of Applied Sciences Göttingen gave their approval to the research protocol.



(a) PHQ-8 scores distribution (b) Age distribution of the participants

Fig. 1. Statistics of KID dataset.

3.4. *Materials*

3.4.1. *Demographic questionnaire*

Prior to filling out the PHQ-9, participants stated their demographic data (age, gender, educational degree and German language literacy) in an online questionnaire.

3.4.2. *Clinical interview for depression*

The GRID-HAMD (Itai et al., 1982) is a depression measure consisting of a scoring system, a manual for scoring conventions and a semi-structured interview guide. It is based on the Hamilton Depression Rating Scale (HDRS) (Hamilton, 1960) and was upgraded to score symptoms of depression on two dimensions: intensity and frequency. Those dimensions form the namesake grid in which numerical values can be attributed to combinations of frequency and intensity. Frequency refers to the relative time in which symptoms are present and is scored on a Likert scale with the increments ‘absent, occasional, much of the time, and almost all of the time’ while intensity refers to the degrees of functional impairment and subjective distress caused by the symptoms. Increments for this axis are rated as ‘absent, mild, moderate, severe, and very severe’ or ‘absent, mild, and marked’. The GRID-HAMD offers significant advantages over the traditional HDRS by providing a clearer differentiation between symptom frequency and intensity. Additionally, GRID-HAMD has improved anchors, standardised evaluation conventions and assessment criteria (Williams et al., 2008), resulting in an overall higher reliability, which can also be achieved by untrained rater (Tabuse et al., 2007). These measures are reflected in an internal consistency of Cronbach’s $\alpha = 0.78$ and an interrater-reliability of $ICC = 0.94$ (Williams et al., 2008). For our experiment, the GRID-HAMD-17 offered the special benefit of an answering format that asked participants to elaborate and therefore produce more free text for our evaluation. Exploratory factor analysis of the HAMD-17 found a four-factor structure with items loading on factors ‘Depression’, ‘Anxiety’, ‘Somatic’, and ‘Insomnia’ (Shafer, 2006). Interviewers worked with the German version of the GRID-HAMD-17 (Schmitt et al., 2015), due to the study taking place in Germany with German-speaking participants. The interview was carried out and recorded via a secure online video conference platform.

3.4.3. *Transcription and translation tool*

To transcribe the interview audio data and translate it from German to English, the speech-to-text model Whisper (OpenAI, 2022) was used. Whisper was specifically developed for automatic speech recognition (ASR) and built upon the foundation of GPT (Generative Pre-trained Transformer) architecture. It has demonstrated unparallelled proficiency in executing transcription and translation tasks for English and German languages with exceptional accuracy, surpassing other existing ASR tools (Seagraves, 2022). The offline and open-source nature of the Whisper model aligns perfectly with the requirements of our application for data sensitivity and data protection.

3.4.4. *Depression screening*

As a baseline measurement of participants’ symptomatic depressive burden, the German version of the 9-item depression scale of the Patient Health Questionnaire (PHQ-9) (Kroenke et al., 2001; Gräfe et al., 2004), was used. The PHQ-9 is a valid, reliable and economic nine question assessment tool for depressive symptoms (Kroenke et al., 2009). Singular items refer to different symptoms of depression and are rated by frequency of occurrence. Participants mark the frequency of symptom occurrence on a Likert scale from 0 (= “not at all”) to 3 (= “nearly every day”). The result can be scored by counting together the score obtained on each of the questionnaire’s 9 items and therefore varies from 0–27. Cut-off values for PHQ-9 indicate none-minimal, mild, moderate, moderately severe and severe depressive symptomatology (Kroenke et al., 2009). Generally, a value of 10 and above is seen as indicative of depression. A high internal consistency, Cronbach’s $\alpha = 0.88$, Gräfe et al. (2004) has been reported for the PHQ-9, which has been validated in multiple studies (Kroenke et al., 2001; Gräfe et al., 2004). A variant of the PHQ-9 is the PHQ-8, which is made up of the same items with item number 9 omitted, as it refers to suicidal ideation Total scores can vary from 0–27, with higher values indicating a higher frequency of depressive symptoms (Kroenke et al., 2009). Due to this it is used for research purposes and was furthermore the Ground Truth value within the DAIC dataset, which was used to fine-tune the BERT-based model within this study. To make our research comparable to the findings of NLP-model performance reported for this dataset (e.g. Arseniev-Koehler et al., 2018; Danner et al., 2023; Morales and Levitan, 2016 and Villatoro-Tello et al., 2021), we decided to also use the PHQ-8 as depression assessment tool.

3.5. Data collection and transformation

The data collection procedure was tested in a preliminary study (Hadžić et al., 2024): Participants registered for the study by contacting the test manager. Specifically appointed intermediaries then got in contact with the participants and guided the assignment procedure by providing them with information regarding the purpose of the study, their individual participant code and a link to an online questionnaire, containing a quick screener on demographic data and the PHQ-9 (Kroenke et al., 2001). Intermediaries inquired whether participants were familiar with any of the interviewers running the study (e.g. from studying at the same campus), before supplying them with a list of time-slots in a web-based anonymous planning tool. Participants could register for a time-slot of their choice in a web-based planning tool. It was ensured that allocation to interviewers who could identify them was blocked. Participants were instructed to neither disclose personal information during interviews nor activate their camera. Interviewers were only provided with participant codes which they could use to access PHQ-9 results from the online survey. Intermediaries, who communicated with participants via email did not have access to any survey data. Additionally, demographic data was collected separately from PHQ-9 scores, ensuring that interviewers were unaware of participant demographics.

Interviewers were psychology students at the Private University of Applied Sciences Göttingen who were trained to conduct the German version of the GRID-HAMD-17 interview (Schmitt et al., 2015) by a clinical psychologist supervised by a psychotherapist. Furthermore, interviewers were instructed to calculate PHQ-9 scores for participants, which would be disclosed at the end of the interview, when participants were provided with information on depression as well as the general scoring conventions of the PHQ-9 which allowed them to classify their score and take the necessary steps to seek treatment in case of high values. The PHQ-9 data was collected in an online-survey. Prior to the interviews, interviewers accessed the online-survey, searched for the dataset associated with the participant code and calculated the PHQ-9 score by adding up the score of the items. Interview audio data was recorded during the interview session, which was carried out on a secure learning platform system. Participants consented to the interview procedure and recording of the audio data before the start of the clinical interview. During the interview, the camera was turned off to guarantee anonymity and no sensitive information which would reveal the participant's identity was collected. The interview followed the GRID-HAMD-17 protocol (Schmitt et al., 2015). The audio files were shortly saved on servers in the European Union (Germany and Ireland) under GDPR regulation, before saving them to private servers on campus, transcribing and deleting them.

In the data preprocessing procedures, transcription and translation from German to English language were carried out using the pre-trained Whisper model large-v2 (OpenAI, 2022), which turned the audio files from the interview (provided in WAV data format) into segments of voice activity. From those segments, transcription of the interview was generated as a spreadsheet in CSV format forming a transcribed dataset of interviews saved in English language. Classification by speaker role ('interviewer' or 'participant') was done manually, as currently available tools were proven to be insufficient. Considering the model's processing limit regarding the maximum size of possibly processed tokens, only the segments spoken by the participant were saved as a string and later given to the models.

Test and fine-tuning datasets for each model were produced and pre-processed according to an identical set of instructions, guaranteeing that each model deals with the exact same assessment conditions, providing same reproducible methods through the evaluation process. Ensuring reliability and equitable comparability among different models was largely dependent on these processes.

3.6. Clustering interview data

To give the LLMs a bit more context while simultaneously executing finer control over their predictions and account for differences in depression measurement between GRID-HAMD-17 and PHQ-8, we introduced clustering of interview data. The clusters were built following the results of Shafer (2006)'s meta-analysis of factor analyses for the HAMD. According to this factor structure, interview text was divided into 5 clusters: depression, anxiety, somatic and insomnia plus a further category named 'unimportant'. The depression cluster contained questions about depressive mood or feelings of guilt, while the anxiety cluster contained questions about mental anxiety symptoms. The somatic cluster contained general physical symptoms like loss of appetite, and the insomnia cluster contained questions about sleep disorders. The decision to create the cluster 'unimportant' was made during the clustering process, when text fragments that fit into none of the categories and did not involve questions and/or answers to the clinical interview, were encountered. Such communication referred mostly to technical problems within the video conference (e.g. "can you hear me?"). Clustering was done by manually labelling questions and answers in the transcript. The cluster 'unimportant' was later deleted, as it did not contain any information pertaining to the clinical interview. We presented the interview text to the model along with its designated cluster category. Finally, we adapted the prompt to get a score between 0 and 6 for each of the 4 clusters and summed them up to get the overall PHQ-8 score. The updated prompt is listed in the Table 1.

An additional incentive to experiment with the clustering strategy for BERT, was to try to obtain a projected score value in the range of 0 to 24, as is possible with other models, given that BERT predictions are always set as binary outputs. Therefore, in the case of the BERT model, the data processing part demanded one additional step. We determined the average score (from binary scores 1 and 0) per cluster and multiplied this result by the cluster weight to obtain scores in the range of 0 to 6. With all four cluster scores considered, this method results in a projected overall score in the range of 0 to 24. With this new approach, we tried to compare the results with the original dataset and other models.

Table 1
An example prompt used for Clustered KID dataset.

Prompt	You are getting the answers of an interview divided into four sections (A: , D: , I: , S:). According to the PHQ-8 scheme, give me a value between 0 and 6 for each section. [Interview] A: Interviewer: I'm going to ask you a battery of questions that all relate to the past seven weeks. Is it okay with you? Participant: Yes Interviewer: How was your mood during the past week? ....
Output:	Based on your responses, you scored as follows in each section of the PHQ-8 scheme: A: 1 (Yes, but that's my state for a few weeks now) D: 2 (I was a bit stressed by the work, but my mood was actually quite good) I: 3 (My usual times of going to sleep are between 10 and 12 in the evening. That's always the tension.) S: 2 (For me, it was inconspicuous. I was hungry and when I was hungry I ate something.)

3.7. Metrics

Recall, also referred to as sensitivity or true positive rate, captures the classifier's efficacy in the identification of all positive cases in the given dataset. The recall value is calculated by dividing the number of positive classifications by the number of total positive cases in the used dataset (Takahashi et al., 2022).
Precision is another relevant metric when it comes to classification tasks. It is focused on the accuracy of the positive prediction that the classifier makes and is calculated by dividing the total number of true positives and false positives by the number of cases that were correctly recognised as positive (true positives). Precision is a crucial indicator that enhances other metrics, such as recall, to give a thorough assessment of a classification model's performance, particularly when the importance or danger of the false positive classification is high (Yacouby and Axman, 2020).
F1 score on the other hand is a combination of introduced precision and recall scores and is always to be considered when both false positive and false negative scores are important to interpret the efficacy of the classifications (Takahashi et al., 2022). It can be mathematically presented through the following formula:

$$F1\text{-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

It is important to note that the scores of these three metrics all range between 0 and 1, where a value that lies closer to 1 indicates superior achievement in capturing true positive scores (recall), accurately predicting positive scores (precision) or striking a balance between the two (F1). To determine ground truth values for calculating introduced metrics, we used PHQ-8 questionnaire scores collected after conducting clinical interviews. Using PHQ-8 scores as the foundation, we provide a thorough evaluation that takes into consideration each error and corrected prediction that the model makes.

The metrics mentioned above are standardised evaluation measures, particularly in the domain of machine learning and NLP. More precisely, these metrics are most commonly used when it is intended to evaluate the performance of the model performing classification tasks. It has proven to be very useful when dealing with imbalanced datasets, as is the case in our approach. A high prevalence of non-depressed participants in the dataset is to be expected in the sample of participants coming from the general population.

Other metrics that we found applicable to get a deeper and better understanding of the performance of the utilised models are Pearson correlation coefficients (PCC), Mean squared error (MSE) and Coefficient of variation (CV). The degree and direction of a linear relationship between two continuous variables can be determined using the Pearson correlation coefficient (PCC). It provides insights into which changes in one variable are connected with changes in another (Saccenti et al., 2020). A value of 0 denotes no linear relationship among the variables, while a value between −1 and 1 indicates a strong linear correlation. Mean absolute error (MSE) is widely used in many various domains for model fitting, model validation, model selection, model comparisons or forecast evaluations (Karunasingha, 2022). They are frequently combined with additional metrics and domain knowledge to offer a thorough assessment of a prediction model (Hodson, 2022). When interpreting the results MSE, there are no predefined criteria for interpreting the significance of the error. Results on these two metrics should be considered as a value of how many score units the model miss-predicted when compared with the ground-truth dataset. When evaluating or comparing models on MSE statistics, higher scores indicate higher prediction error and therefore, more inferior performance. In comparison to the Precision, Recall and F1 scores, MSE provide more detailed feedback on the performance of the model, considering the exact offset of predicted score from the target (ground truth) value. On the other hand, Precision, Recall and F1 score are considering how successful are the binary outputs made by the model considering proportions of true and false predictions. Together, these metrics provide thorough assessment of the evaluated models. Lastly, to measure the variability and dispersion of the predicted values, we also considered the Coefficient of Variation (CV). The coefficient of variation (CV) is a dimensionless statistical metric used to estimate, as a percentage, the relative dispersion or variability of a dataset, whereas higher values mean greater variability compared to the mean (Abdi, 2010). Considering its dimensionless characteristic, it is a very common metric to use when comparing variability and dispersion of results collected on different scales or sample sizes.

Table 2
Performance Metrics on original datasets.

Work	Precision	Recall	F1 score
Preliminary KID Dataset n = 17			
BERT (Previous work (Hadžić et al., 2024)	0.69	0.71	0.70
GPT-3.5 (Previous work (Hadžić et al., 2024)	0.92	0.88	0.89
KID Dataset n = 82 (Current Study)			
BERT (Fine-tuned)	0.66	0.71	0.66
GPT-3.5 (Zero-shot)	0.82	0.36	0.34
GPT-4 (Zero-shot)	0.78	0.72	0.73
Llama2-13B (Zero-shot)	0.63	0.45	0.47

Table 3
Performance Metrics on Clustered KID Dataset.

Work	Precision	Recall	F1 score
BERT (Fine-Tuned)	0.61	0.71	0.61
GPT-3.5 (Zero-shot)	0.78	0.66	0.68
GPT-3.5 (Fine-tuned)	0.85	0.84	0.82
GPT-4 (Zero-shot)	0.75	0.76	0.76
Llama2-13B (Zero-shot)	0.72	0.66	0.67

4. Results

In order to assess the effectiveness of the models used in our approach, we used the Precision, Recall and F1 score metrics as the starting point of our model evaluation methodology. Table 2 presents the outcomes of the conducted analysis. In comparison to the previous study, a significant increase in the volume of data was observed. Consequently, this resulted in a notable decrease in the performance of the GPT-3.5 model and a marginal decline in the performance of BERT. While GPT-4 demonstrated an accuracy of 0.73 at zero-shot, the open-source Llama3-13B model performed the task with a lower accuracy of 0.47.

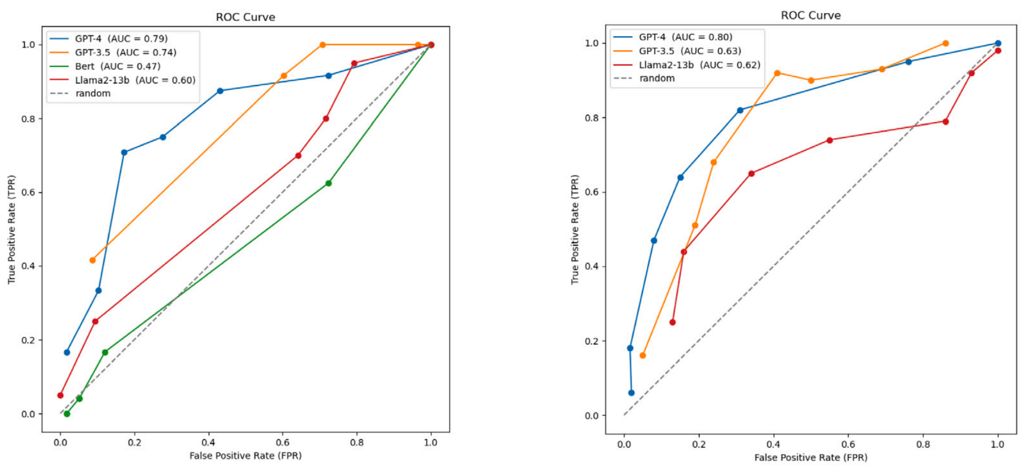
To assess the efficacy of the proposed clustering idea, we performed predictions on our Clustered KID Dataset using GPT-3.5 and Llama2-13B models in order to test if clustering improves the raw out-of-the-box generalisation capabilities of these models. Specifically, to test if clustering would encourage the GPT-3.5 model to predict higher PHQ-8 scores. Table 3 shows that clustering has considerably improved the metrics both in terms of Recall and F1 score, which is highly relevant for our application. This clearly shows that providing the model with more context and finer details about the interviews increases the likelihood of better predictions. The BERT model, however, did not show any improvement.

To further improve the results, we optimised the classification threshold on a per-model basis using a ROC/AUC method. Fig. 2(a) shows ROC curves for all our models. A ROC curve visually depicts the trade-off between correctly classifying positive cases (True Positive Rate) and incorrectly classifying negative cases as positive (False Positive Rate) at various decision thresholds. An optimal threshold is chosen by aiming for a high True Positive Rate while keeping the False Positive Rate low. This resulted in selecting a new optimal threshold of T*=12 for all models, viz. GPT-3.5, GPT-4 and Llama2-13B with an improved F1 score of 0.67, 0.75 and 0.69 respectively. This experiment was then redone with the new clustered KID dataset, and the resulting ROC curve is shown in Fig. 2(b). However, this time the optimal threshold remained the same at T*=10.

Finally, to correct the GPT-3.5 model's tendency to predict lower PHQ-8 scores and improve its performance in general, we decided to fine-tune it. We selected the top 12 interview samples of the participants with the highest error in the PHQ-8 scores as training data for fine-tuning tasks. The JSON file used for this has examples with the interview text as input and the PHQ-8 score as output. This fine-tuning step was performed using the clustered KID dataset introduced the previous section. Fine-tuning the GPT-3.5 model as anticipated led to a notable enhancement in its performance, as shown in the Table 3.

To account for the prediction of symptom severity done by the models, further statistical comparisons were calculated. The Table 4 presents a comparison of models and approaches, including GPT-3.5's performance in standard form with standard data, clustered data, and the fine-tuned version, GPT-4 in standard form with standard data and also with clustered data, Llama2-13B in its standard form as well as with the clustering approach. From this comparison, we left out the BERT standard model, as its classification output is binary in nature and is therefore not comparable with the other models on selected statistic metrics. However, clustering the dataset into smaller segments did allows us to obtain results comparable with other models, regarding these specific metrics. The evaluation metrics include Mean (M), Standard deviation (SD), Pearson correlation coefficient (PCC), Mean square error (MSE), and Coefficient of variation (CV). Results of conducted statistical tests are presented in Table 4. In the last row, for comparison purposes, the table includes statistics for the true data depression score measured on the PHQ-8 questionnaire.

Beginning with the mean (M) values, we observe that GPT-3.5 Fine-tuned exhibits the lowest mean score (M = 6.30), suggesting that, in comparison to other models, it tends to give prediction values that are lower than the true data set mean. This tendency might be due to a higher proportion of lower depression scores in the dataset used for fine-tuning. GPT-3.5 Clustered, on the other hand, has the highest mean value (M = 9.71), indicating that it generally tends to predict higher scores. With a standard deviation (SD) of SD = 6.22, GPT-4 has the highest degree of variability across predictions when compared to other models, suggesting greater



(a) Original KID dataset. (b) Clustered dataset.

Fig. 2. ROC Curves showing the performance of various models on (a) Original KID dataset vs on (b) Clustered KID dataset.

Table 4
Further statistical comparison.

Model	M	SD	PCC ↑	MSE ↓	CV
BERT (Clustered)	5.14	2.735	0.058	34.012	53.22%
GPT-3.5	9.51	2.736	0.548**	18.451	28.77%
GPT-3.5 (Clustered)	9,71	3.633	0.465**	21.314	37.45%
GPT-3.5 (Fine-tuned Clustered)	6.30	4.001	0.658**	13.143	63.51%
GPT-4	9.07	6.226	0.706**	21.280	68.64%
GPT-4 (Clustered)	8.67	5.220	0.524**	26.971	60.20%
Llama2-13B	8.33	4.038	0.248*	28.438	48.48%
Llama2-13B (Clustered)	8.85	4.385	0.266*	30.971	49.54%
True Data Stats	7.65	4.649	–	–	60.77%

variation in the range of predicted values. In contrast, GPT-3.5 has the lowest standard deviation (SD = 2.74), indicating that its predictions are more closely distributed around the mean. This low variability, noticed in the standard GPT-3.5 model, is improved with clustering and fine-tuning since SD and CV scores in these approaches increase and get closer to the SD observed in the actual data. With a Pearson correlation coefficient (PCC) of r = 0.71, GPT-4 has the strongest correlation with the true data, suggesting a robust positive linear relationship, while PCC of clustered GPT-4 drops to r = 0.53. GPT-3.5 Fine-tuned, with a PCC of r = 0.66, comes a close second. Llama2-13B, on the other hand, has the second lowest PCC value with r = 0.25 at zero-shot. Clustering led to a minimal improvement for Llama2-13B which suggests a very weak linear correlation with the true data. Lowest PCC values are measured with BERT clustered dataset, r = 0.06 which implies that there is no correlation between the two distributions. GPT-3.5 Fine-tuned exhibits the lowest mean squared error (MSE) at 13.14, suggesting superior predictive accuracy, while BERT Clustered exhibits the highest MSE at 34.01, indicating poor predictive performance when compared to the other models.

5. Discussion

In the present study, the efficacy of four distinct Large Language Models (LLM) in identifying depression from clinical interviews was assessed, simulating a potential real-world application. The models evaluated included zero-shot trained Llama2-13B, GPT-3.5, GPT-4, and a pre-trained BERT-based model. Among these, GPT-4 demonstrated superior performance with an F1 score of 0.73, followed closely by the BERT model at an F1 score of 0.66. While GPT-3.5 performed with a high accuracy of 0.89 in the preliminary dataset (Hadžić et al., 2024), accuracy within the bigger dataset of this study dropped to 0.34. Upon closer inspection, it is revealed that this phenomenon stemmed from the conservative nature of the untrained (zero-shot) GPT model, which tended to assign lower PHQ-8 scores within the range of 7–10. Initially, this tendency proved effective given the limited dataset with fewer depressed participants. However, with the expansion of the dataset, this approach became less viable. This tendency is reflected in precision and recall metrics: Precision metrics for all models were consistently high (≥ 0.63). However, the untrained versions of GPT-3.5 and Llama2-13B exhibited lower Recall values (≤ 0.45), indicating a tendency to underestimate depression prevalence within the study population. While GPT-3.5 demonstrated largely different values for Precision and Recall, Llama2-13B performed worse in all metrics, which can be attributed to several factors, including the absence of depression interview data during pre-training, limited generalisation capacity, or the potential need for additional contextual information.

To enhance model performance, we explored two strategies: clustering the data and fine-tuning the models. These interventions significantly improved Recall for GPT-3.5, achieving scores of 0.66 with clustering and 0.84 with fine-tuning. It was observed that while clustering slightly reduced the Precision by 0.04 in GPT-3.5, fine-tuning resulted in an increase of approximately 0.03 in Precision. Consequently, the overall F1 score for GPT-3.5 was elevated from 0.34 to 0.68 with clustering and further to 0.82 with fine-tuning, surpassing the F1 score of the zero-shot GPT-4 model. For GPT-4, our clustering approach led to an improvement of the F1 score by 0.03 Overall, clustering resulted in a substantial improvement in F1 scores for the GPT-3.5, GPT-4 and Llama2-13B models but led to decrease of F1 score of BERT model for 0.05. This is due to the fact that BERT has a limited context window length. Therefore, adding more context does not improve model performance. These enhancements suggest that methods such as few-shot learning and clustering can improve model performance with regard to F1-scores.

Furthermore, to assess the models' capability to accurately estimate the severity of depressive symptoms, Pearson correlation coefficients were calculated. The zero-shot GPT-4 model exhibited high correlation ($r = 0.71$) with self-reported measures of depression severity, outperforming the fine-tuned GPT-3.5 model, which, despite achieving a higher F1 score, demonstrated less accuracy in evaluating the severity of depression. For GPT-3.5 and GPT-4, clustering the data decreased correlation of model predictions with actual PHQ-8 scores by 0.08 (for GPT-3.5) and 0.18 (for GPT-4). For Llama2-13B, clustering the dataset improved model performance by 0.02. These findings underscore the nuanced capabilities of these models, not only in detecting the presence of depression but also in assessing its severity. Furthermore, the lower correlations obtained from a clustered dataset for GPT-3.5 and GPT-4 shed light on the importance of testing for different model performance metrics when evaluating a new approach.

Regarding the MSE, results showcase that the GPT-3.5 Fine-tuned model achieved superior predictive accuracy and consistency compared to other models due to its low mean square error overall. These findings underscore the importance of model fine-tuning for specific tasks.

In contrast to the methodologies used in similar studies, as outlined in recent reviews by Ahmed et al. (2022) and Zhang et al. (2022), our study employs a sample from the general population. This approach enhances the external validity of our results by being reflective of the spectrum of depression severity encountered within the general population. Such inclusivity is critical for the real-world applicability of automated depression screening tools, as it mirrors the diversity of individuals these tools are likely to encounter in practical settings.

Llama2-13B, being an open-source model, does not reach the performance levels of closed-source GPT models. It does, however, show a great deal of potential, especially with the possibility of fine-tuning processes to adjust model specifically for depression detection tasks. At the same time, data security, privacy, and transparency are given high priority in open-source models like Llama2. These are very important considerations for managing sensitive data, such as user mental health records.

The accuracy scores for BERT within our current dataset did not match those of the preliminary dataset (Hadžić et al., 2024) as presented in Table 2, though they still surpassed the benchmarks set by Villatoro-Tello et al. (2021) and Senn et al. (2022). A reason for the less favourable results in comparison with our previous study (Hadžić et al., 2024) is the higher number of depressed individuals within this sample combined with BERT's lower recall values. This variation highlights the challenge of achieving consistent model performance across different data sets, underlining the importance of testing predictive models within a sample containing several cases distributed around the cut-off value.

GPT-4 demonstrated substantial accuracy in predicting PHQ-8 scores from semi-clinical interview transcripts, aligning with the successful application of speech processing and LLM-based text summarisation in depression detection as reported by Sadeghi et al. (2023). Conversely, while Lamichhane (2023) found GPT-3.5 to achieve an F1 score of 0.86 in detecting depression from social media posts, our findings showed a much lower F1 score of 0.34 for zero-shot GPT-3.5 (which improved after fine-tuning), suggesting that results obtained in the evaluation of social media content may not carry over to clinical interview data.

The superior performance of GPT-4 (F1 score of 0.73), can be attributed to its training on a significantly larger dataset, enhancing its inferencing capabilities as discussed in Zhong et al. (2023). GPT-4's ability to interpret symptoms and correlate them with PHQ-8 criteria suggests inferencing capabilities of depressive indicators, underscoring the potential of advanced NLP models in accurately detecting depression across various contexts. Still, GPT-4 results do not reach the Specificity and Recall values obtained by Victor et al. (2019)'s deep learning approach, which might be due to interview questions directly referring to a history of depression or due to the superiority of an integration of multimodal data.

To our knowledge, our study is pioneering research on GPT-4 capacities in depression detection, adding a novel dimension to the growing body of literature exploring the potential of advanced language models in mental health applications.

5.1. Limitations

While this study aimed to discern whether NLP models can detect depression, it is important to notice that different criteria for depression are posed by the GRID-HAMD-17 and the PHQ-8. This can be derived from a convergent validity of $r = 0.66$ (Schmitt et al., 2015) between HAMD and PHQ-9. Given this score, the correlation obtained by GPT-4 ($r = 0.71$) appears quite proficient. It is important to notice however, that the model was prompted to infer a PHQ-8 value and not score the GRID-HAMD-17. Whether the NLP models ability to score an interview approaches that of a human rater should be investigated in further research. The potential to mitigate differences in depression criteria by segmenting interview transcripts into factors following the structure identified by Shafer (2006)'s meta-analysis was explored. Model prompts were adapted accordingly. While a relatively small increase in performance was found for Llama2-13B, GPT-3.5, GPT-4 and BERT did not profit in accuracy from clustering approach and instead showed worse performance in the prediction of symptom strength. Future studies are necessary to refine these methods or explore additional ways to harmonise disparate data sources.

The absence of non-verbal and paraverbal cues in our analysis, such as visual signs of psychomotor symptoms or voice modulation, is another limitation. Recognising the importance of these cues in comprehensive depression assessment, we suggest that subsequent research could incorporate multimodal data analysis, as exemplified by Victor et al. (2019), to enrich the dataset. Technologies enabling the integration of visual and auditory data could offer a more holistic understanding of depressive symptoms.

The transcription process itself was a potential source of errors within our study. While one file got corrupted within the transcription process so that it showed multiple repetitions of single words or phrases, the usual amount of transcription errors was relatively low. Except for the removal of the corrupted file, we decided against manually correcting transcription errors, since an automated transcription process would likely be part of an application in a screening app and correcting for this would negatively impact the external validity of our findings.

Speaker annotation appeared as a challenge within our research and could not be carried out as an automated task within the transcription process. It was therefore done manually, which can be seen as a limitation to the external validity of our study.

Translation inaccuracies from German to English presented challenges in our approach. In some cases, the translation changed the original meaning of a segment, which was not corrected, with regard to external validity. Translation in general might have led to some loss of linguistic nuance. Future studies might employ more sophisticated translation processes or bilingual analysis to preserve the subtleties of language, which is crucial given the significance of linguistic patterns in diagnosing depression (Wolohan et al., 2018). Furthermore, a comparison with studies of English native speakers which did not need translation, would be interesting.

Regarding sample composition, we aimed at attaining a more diverse sample by diversifying recruitment strategies: Through word-of-mouth from our students and the course-credit board, we attempted to counteract a self-selection bias as we might have encountered in a sample recruited from one platform only (e.g. social media). At the same time, we aimed at including members of the general public by not relying on the universities' course credit board alone. While the age range within our sample shows that people who are not the typical student population were included, the peak in frequency around the ages 20–30 shows there is still much room for improvement regarding sample diversity. Furthermore, our study featured a higher proportion of female participants. It is highly suggested to take recruitment strategies beyond the measures taken by us to ensure a diverse sample and test NLP models for bias in determining depression scores for marginalised and/or underrepresented groups. Future efforts could aim to diversify recruitment strategies to ensure broader representativeness, thereby enhancing the applicability of findings across different populations.

Lastly, the cultural and linguistic diversity within our sample was limited. Recognising the influence of cultural norms on the presentation and perception of depressive symptoms (Lehti et al., 2010), as well as on responses to assessment instruments (Iwata and Buka, 2002), we advocate for the inclusion of culturally diverse populations in future studies. This approach would not only enrich the dataset but also enable the examination of NLP models' ability to navigate cultural and linguistic variations in depression diagnosis, moving towards more equitable and universally applicable diagnostic tools.

5.2. Future directions

We focused on examining the effectiveness of different NLP models in analysing clinical interviews by giving estimations of participants' depression severity self-report values. Further research is needed comparing NLP model performance to human evaluators. This exploration is intended to further understand the capabilities and limitations of AI models in clinical contexts, offering insights that could enhance their accuracy and reliability. Addressing the need for diverse and inclusive research samples is another area where our study seeks to add value. By advocating for the inclusion of participants from a wide range of demographic backgrounds, we aim to support the development of AI systems that are equitable and effective for a broader population. This effort aligns with the broader research community's goal to mitigate biases within AI applications in mental health. Exploring the integration of multichannel data analysis (e.g. Victor et al., 2019) represents an extension of current methodologies, rather than a novel departure. Incorporating additional data types, such as prosodic and visual cues, into depression detection models is a promising research direction for improving accuracy and gaining insight into depression biomarkers and their presentation in different domains. Our approach to addressing the ethical challenges associated with using online data for depression detection reflects a commitment to privacy and data protection. By opting for semi-clinical interviews as a data source and utilising offline, open-source language models, we contribute to the ongoing discussion about ethical standards in AI research, emphasising the importance of safeguarding mental health information.

5.2.1. Considerations related to automation

In the context of automating depression detection, the challenges we identified – such as speaker annotation, transcription accuracy, and managing input token limits – highlight areas for continued improvement. These operational issues shed light on some of the complexities involved in transitioning towards fully automated systems.

Currently, the process of clustering interview data is conducted manually by reviewing segments of the interview transcript and assigning one of five predetermined labels. This method is labour-intensive and time-consuming. In future endeavours, we propose implementing a sentence clustering algorithm to automatically categorise interview segments into predefined clusters.

5.3. Practical implications

One potential application of our findings is the creation of an app-based interview system. This system could use NLP models to conduct (semi)clinical interviews, providing a user-friendly interface that encourages individuals to engage with mental health services. Such an application addresses critical barriers in the current diagnostic process, including the stigma associated with mental health issues. An AI-based tool could significantly reduce reluctance to seek help for mental health concerns, as barriers to help-seeking, like resource scarcity (Colligan et al., 2020), perceived stigma (Arnaez et al., 2020) and relational factors (Kravitz et al., 2011) can potentially be overcome by AI-based depression screening systems which offer accessibility, constant availability, anonymity and objectivity. Moreover, the accessibility and cost-effectiveness of an app-based tool present a viable solution for individuals whose conditions limit their ability to seek traditional forms of help. This approach not only broadens the reach of mental health services but also contributes to democratising access to care. However, the neutrality of AI models regarding socio-demographic factors, while potentially reducing human bias, introduces the need for careful consideration of how such factors influence depression detection. The absence of bias related to income, ethnicity, and gender is a double-edged sword, necessitating further research to ensure these models do not inadvertently overlook the nuanced ways in which these factors impact mental health. Further limitations regarding the use of NLP-models for depression detection concern the modalities not covered by natural language understanding: A unimodal approach relying solely on textual data is unable to take in cues from voice, facial expression, body language or appearance — cues that a clinician following psychopathological assessment according to the Association for Methodology and Documentation in psychiatry (AMDP) system (Stieglitz et al., 2017) takes into consideration. Furthermore, the verbal expression of emotions and thoughts differs between cultures, which could potentially contribute to the model under- or overestimating depression scores for certain cultures (Kessler and Bromet, 2013). Similarly, correctly identifying the various manifestations of depression from interview data and differentiating them from other mental health concerns (e.g. bipolar disorder or the prodromal phase in schizophrenia), non-clinical emotional states (like grief), neurological disorders (e.g. dementia) or primarily somatic disorders, is an important task within the diagnostic process (Perrotta, 2019) for which NLP models have yet to be tested. Data on the identification and differential diagnosis of non-affective psychosis (Ferrara et al., 2022) by machine learning algorithms already give a positive lookout, yet more research is needed. NLP models can assist clinical experts in these tasks, but not replace them. In order to be of help, these tools must be easily interpretable and comprehensible for experts. Additionally, it is crucial for users to have initial access to the technology. Lastly, the handling of data in the field of mental health raises ethical concerns regarding data security guidelines (Nanomi Arachchige et al., 2021; Zhang et al., 2022).

6. Conclusion

In conclusion, our investigation into the application of LLMs for depression detection underscores the transformative potential of AI in mental health diagnostics. Despite the promise shown by AI tools, the journey towards their full realisation in clinical practice is still a challenge. Challenges related to demographic representation, potential biases, and the intricacies of automation underscore the critical need for continued research. Future efforts must focus on addressing these challenges to ensure the ethical and inclusive deployment of AI in mental health care, ultimately enhancing the efficacy, accessibility, and acceptability of depression screening and diagnosis.

CRediT authorship contribution statement

Julia Ohse: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Bakir Hadžić:** Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Parvez Mohammed:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Formal analysis, Data curation. **Nicolina Peperkorn:** Writing – review & editing, Project administration, Methodology. **Michael Danner:** Validation, Methodology, Formal analysis, Data curation. **Akihiro Yorita:** Conceptualization. **Naoyuki Kubota:** Supervision. **Matthias Rätsch:** Supervision. **Youssef Shiban:** Visualization, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Bakir Hadžić, Mohammed Parvez, Michael Danner, Matthias Raetsch reports financial support was provided by VDI VDE Innovation und Technik GmbH. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Declaration of Generative AI and AI-assisted technologies in the writing process

Statement: During the preparation of this work the authors used ChatGPT (GPT-3.5/GPT-4), DeepL and QuillBot in order to improve language. After using these tools/services, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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